PARISHRAM 2.0 2026 (Class 12th)

Mathematics

DPP: 05

Relations and Functions

Q1 A function f(x) is said to be one-one if
(A) $f(x_1) = f(x_2) \Rightarrow -x_1 = x_2$
(B) $f(-x_1) = f(-x_2) \Rightarrow -x_1 = x_2$
(C) $f(x_1) = f(x_2) \Rightarrow x_1 = x_2$
(D) None of these

Q2 f: X⟶ Y is onto, if and only if:
(A) $\text{range of } f = Y$
(B) $\text{range of } f \subset Y$
(C) $\text{range of } f \neq Y$
(D) $\text{range of } f \supset Y$

Q3 If X = {4, 5, 6} and Y = {0, 1} and f : X ⟶ Y defined by
f = {(4, 1), (5,1), (6, 0)}, then f is
(A) one-one, into
(B) one-one, onto
(C) many-one, into
(D) many- one, onto

Q4 Set A = {a, b, c, d} and B = {5, 6, 8, – 2} and let a function defined from set A to set B, f = {(a, 6), (b, 8), (c, –2), (d, 5)}, then f is
(A) Many-One
(B) One-One Onto
(C) One-One Into
(D) Many-One Onto

Q5 Let A = {1, 2, 3}, B = {4, 5, 6, 7} and let f = {(1, 4), (2,5), (3, 6)} be a function from A to B. Function f is
(A) one-one and onto
(B) many- one and onto
(C) many-one and into
(D) one-one and into

Q6 The function f : N ⟶ N, f (x) = 2x is
(A) one-one and onto
(B) one-one and into
(C) many- one and onto
(D) many-one and into

Q7 Function f : N ⟶ N, f(x) = 2x + 3 is
(A) One-one onto
(B) One-one into
(C) Many-one onto
(D) Many-one into

Q8 Set A has 3 elements and set B has 4 elements. Then the number of injective mappings that can be defined from A to B is
(A) 144 (B) 12
(C) 24 (D) 64

Q9 The function f : R ⟶ R, f(x) = |x| is
(A) one-one and into
(B) one-one and onto
(C) many-one and into
(D) many-one and onto

Q10 The greatest integer function f : R ⟶ R, given by f(x) = [x] is
(A) one-one
(B) onto
(C) both one-one and onto
(D) neither one-one nor onto

Answer Key

Q1	C
Q2	A
Q3	D
Q4	B
Q5	D
Q6	B
Q7	B
Q8	C
Q9	C
Q10	D

Hints & Solutions

Note: scan the QR code to watch video solution

Q1 Text Solution:

A function f(x) is said to be one-one if

$f(x_1) = f(x_2) \Rightarrow x_1 = x_2$

i.e. for every image there is unique pre-image.

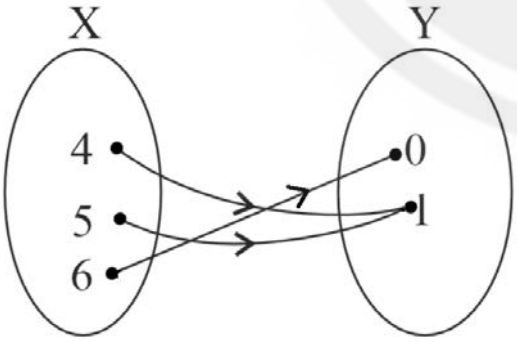
Video Solution:

Q2 Text Solution:

A function $f: A \rightarrow B$ is said to be onto, if for every $b \in B$, there exists an element a in A such that f(a) = b.

Video Solution:

Q3 Text Solution:



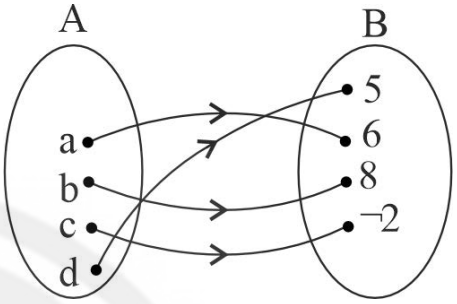
As for element $1 \in Y$ there are two pre-images

$\therefore$ f is many one function i.e. not one-one

As for both elements in Y there is a pre-image in X $\therefore$ f is onto function

Video Solution:

Q4 Text Solution:

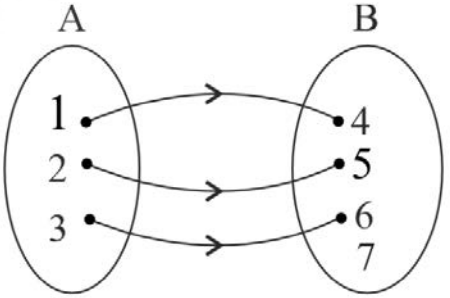


As for every image in B there is unique pre-image in A $\therefore$ f is one-one.

As for all element in B there is pre-image in A $\therefore$ f is onto.

Video Solution:

Q5 Text Solution:



As for every image in B there is unique pre-image

in A $\therefore$ f is one-one

As $7 \in B$ does not have any pre-image in set A

$\therefore$ f is not onto

Video Solution:



Q6 Text Solution:

One-one

Let $x_1, x_2 \in N$

such that $f(x_1) = f(x_2)$

$\Rightarrow 2x_1 = 2x_2$

$\Rightarrow x_1 = x_2$

$\therefore$ f is one-one function

Onto

f is not onto i.e. into function

$\because$ for 3∈N(co-domain) there does not exist an element x∈N (Domain) such that f(x) = 3

$\therefore$ f is into function

Video Solution:



Q7 Text Solution:

One-One

Let $x_1, x_2 \in N$

such that $f(x_1) = f(x_2)$

$\Rightarrow 2x_1 + 3 = 2x_2 + 3$

$\Rightarrow 2x_1 = 2x_2$

$\Rightarrow x_1 = x_2$

$\therefore$ f is one-one

Onto

f is not onto i.e. into function

$\because$ 3∈N(Co-domain) there doesnot exists x∈N(Domain) such that f(x) = 3

i.e. there is no pre-image for 3∈N (Co-domain)

Video Solution:



Q8 Text Solution:

The total number of injective mappings from the set containing 3 elements into the set containing 4 elements is ${}^4P_3 = 4! = 24$.

Video Solution:



Q9 Text Solution:

f(x) is many one function

$\because$ f(1) = f(–1) = 1

but – 1 ≠ 1

Now, f(x) is into function

As there is no element in x ∈ R

such that f(x) = –1

i.e. no pre-image for 1∈R (co-domain)

Video Solution:



Q10 Text Solution:

We know

[2.2] = 2

[2.5] = 2

$\because$ [2.2] = [2.5]

but 2.2 ≠ 2.5

⇒f is not one-one function.

Now, consider 0.6 ∈ R

It is know that f(x) = [x] is always integer. Thus, there does not exist any element x∈R such

that f(x) = 0.6

$\therefore$ f is not onto.

Hence, Greatest integer function is neither one-one nor onto.

Video Solution:



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